Fiscal Year:	FY 2019	**Task Last Updated:**	FY 07/24/2025
PI Name:	Mignot, Emmanuel M.D., Ph.D.		
Project Title:	HERO Twin Astronaut Study Consortium (TASC): Immunome Changes in Space		
Division Name:	Human Research		
Program/Discipline:			
Program/Discipline--Element/Subdiscipline:	HUMAN RESEARCH--Biomedical countermeasures		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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City:	Palo Alto	**State:**	CA
Zip Code:	94304-1334	**Congressional District:**	18
Comments:			
Project Type:	Flight	**Solicitation / Funding Source:**	2013-14 HERO NNJ13ZSA002N-TWINS
Start Date:	04/14/2014	**End Date:**	10/13/2018
No. of Post Docs:	0	**No. of PhD Degrees:**	0
No. of PhD Candidates:	0	**No. of Master' Degrees:**	0
No. of Master's Candidates:	0	**No. of Bachelor's Degrees:**	0
No. of Bachelor's Candidates:	0	**Monitoring Center:**	NASA JSC
Contact Monitor:		**Contact Phone:**	
Contact Email:			
Flight Program:	ISS		
Flight Assignment:	ISS NOTE: End date should be 10/13/2018 per NSSC information (Ed., 7/16/19) NOTE: Extended to 4/13/2019 per NSSC information (Ed., 10/16/18) NOTE: Extended to 10/13/2018 per NSSC information (Ed., 6/20/18) NOTE: Extended to 4/13/2018 per PI and NSSC information (Ed., 4/21/17)		
Key Personnel Changes/Previous PI:			
COI Name (Institution):	Davis, Mark Ph.D. (Stanford University) Snyder, Michael Ph.D. (Stanford University)		
Grant/Contract No.:	NNX14AH25G		
Performance Goal No.:			
Performance Goal Text:			

Task Description:	In this project, the Mignot, Davis, and Snyder laboratories will work at characterizing personalized changes in baseline immune abnormalities and stimulated immune response of astronauts Mark and Scott Kelly in space after flu vaccination. Mark and Scott Kelly's immune data will be studied in parallel before launch (October 2014), following launch, toward the middle of the flight (October 2015), and following return to Earth (October 2016). This level of immune characterization is unprecedented and will be tailored to Mark and Scott's particular immune genetic background (identical in both twins, but different from most other individuals). In order to do this, we will collect baseline time-matched serum and Peripheral Blood Mononuclear Cells (PBMCs) from Scott and Mark Kelly, and measure the response of the same immune parameters to a seasonal trivalent flu vaccination performed at usual yearly schedules during prolonged space travel (October 2015) in comparison to before and after travel (October 2014 and 2016). Vaccination responses will be measured 7 days after baseline sampling of aim 1 at October time points, an ideal time point where B and T cells responses are maximized. Furthermore, it will be determined whether immunologic changes are associated with significant events and physiological changes occurring during the space mission, as determined by other projects. Recent advances in technology, especially within the Stanford Immunology Institute (CCHI), have created opportunities that did not exist even a few years ago. In particular, cytometry by Time-Of-Flight or CyTOF technology, an extension of flow cytometry in which antibodies are tagged with isotopically pure rare Earth elements (Nolan), and the advent of high throughput sequencing technology, particularly of single T and B cells (Mignot, Quake, Robinson, Davis at Stanford) and single cell RNA SEQ (Quake) open up a myriad of possibilities with small volume samples that we particularly wish to exploit in this project. In addition, we now have a large and growing library of human leukocyte antigen (HLA) alleles as tetramers. Tetramers are used to detect and quantify T cells that are specific for a given antigen within a blood sample. With use of increasingly strong HLA binding predicting tools (such as netMHC3.0 epitope prediction software), it is basically now possible to create tetramer probing T cells for any influenza epitope in any given individual. Interestingly, these HLA-epitope combinations are also promiscuous, so that they may detect T cells that can cross-react with other pathogen epitopes. The establishment of a permanent human presence aboard the International Space Station (ISS) is one of our greatest human triumphs and paves the way for future manned explorations of our solar system. The rigors of space travel present conditions for which our species have not adapted. This project aims at studying the nature and magnitude of immune dysfunction during extended space travel in one subject, with the goal of mitigating it one day in other subjects, possibly through the use of vaccination.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	A very atypical change in cellular subset or gene expression could similarly lead to some new hypothesis and models through the study of literature. Additionally, a better understanding of space-induced immune dysregulation specific to the astronaut, and likely generalizable to others, will help design more targeted future experiments for long-term space travelers.
Task Progress:	Final reporting not received. See Bibliography for resulting publications since last report (Ed., 7/24/25).
Bibliography Type:	Description: (Last Updated:)